

Intermediate Elective 2

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1. Explaining my Inspiration

I chose Steven Ponces' "Coal falls. Renewables rise" as my visualization template because I liked how simplistic and sleek his representaiton of energy data is. This style makes sense for the data I plan to present, because it charts 2 variables (coal vs. renewable energy) that use the same units over the same time in history. I plan to chart % coverage of algae and % coverage of coyote-thistle in the same fashion, to see how the the two variables follow or seperate from each other. From the veg dataset, I'll be using num_quad and percent_cover to calculate a mean_cover for both algae and coyote-thistle, which will be on my y-axis, and I'll use the date column for the x-axis to show change over time.

2. Sketching my figure

A pdf of the sketch of my figure can be found [here](#)

3. Coding up the Figure Using Ponce's Template

1. Packages

```
## 1. LOAD PACKAGES & SETUP ----
suppressPackageStartupMessages({
  pacman::p_load(
    tidyverse, ggtext, showtext,
    janitor, scales, glue,
    camcorder, here, magick
  )
})
```

```
})
```

2. Data

```
veg <- read_csv(here("data", "veg.csv"))
```

```
Rows: 27036 Columns: 17
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr  (10): monitors, transect_name, vp_axis, site, transect_side, vp_zone, q...
```

```
dbl  (6): x1, transect_length, num_quad, transect_distance, percent_cover, ...
```

```
date  (1): date
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
metadata <- read_csv(here("data", "vp_veg_metadata.csv"))
```

```
Rows: 257 Columns: 4
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (2): year_pool, transect_name
```

```
dbl (2): year, num_quad
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Necessary Code from Class

```
# creating new object from veg
pools_explicit0 <- veg %>%
  # filtering for vernal pools at ncos sites
  filter(str_detect(transect_name, "VP") &
         site == "ncos") %>%
  # handling missing data and filling with 0 values
  complete(year, transect_name, psoc, transect_distance,
```

```

    fill = list(percent_cover = 0)) %>%
# grouping by year, transect_name, and species
group_by(year, transect_name, psoc) %>%
# calculating percent cover per year
summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) %>%
#ungrouping the df
ungroup() %>%
# joining with metadata to get year_pool
unite("year_pool", year, transect_name, na.rm = TRUE) %>%
left_join(metadata, by = "year_pool") %>%
# making a new column of mean percent cover
mutate(mean_pc = sum_pc/num_quad) %>%
# grouping by year and species
group_by(year, psoc) %>%
# calculating mean percent cover per year across pools
summarize(mean_pc_year = mean(mean_pc, na.rm = TRUE)) %>%
#ungrouping
ungroup()

```

`summarise()` has grouped output by 'year', 'transect_name'. You can override using the `.groups` argument.

`summarise()` has grouped output by 'year'. You can override using the `.groups` argument.

```

# filtering to investigate only coyote-thistle and algae
filtered_explicit0 <- pools_explicit0 %>%
  filter(psoc == "Eryngium_vaseyi" | psoc == "algae")

```

3. Tidy Data

```

# retrieving the peak value of algae cover
algae_peak <- filtered_explicit0 %>%
  filter(psoc == "algae") %>%
  slice_max(mean_pc_year)

# defining the year when algae peaked
algae_peak_year <- algae_peak$year
# defining the value of peak algae
algae_peak_val <- round(algae_peak$mean_pc_year, 1)

```

4. Visualization Parameters

```
### /- plot aesthetics ----
colors <- list(palette = list(
  bg_color      = "#0D1B2A",
  coyote_gold   = "#F4A932",
  algae_green   = "#009021",
  text_white    = "#FFFFFF",
  text_muted    = "#8A98A8",
  text_sub      = "#B0B8C4",
  grid_color    = "#1A2E42",
  accent_coral  = "#E8503A"
))

# assigning colors to objects
bg_color      <- colors$palette$bg_color
coyote_gold    <- colors$palette$coyote_gold
algae_green    <- colors$palette$algae_green
text_white     <- colors$palette$text_white
text_muted     <- colors$palette$text_muted
text_sub       <- colors$palette$text_sub
grid_color     <- colors$palette$grid_color
accent_coral   <- colors$palette$accent_coral
col_coyote     <- coyote_gold
col_algae      <- algae_green

### /- titles and caption ----
# writing a snappy title
title_text <- glue(
  "<span style='font-weight:900; color:{algae_green};'>Algae
    ↪ Falls.&nbsp;</span>",
  "<span style='font-weight:900; color:{coyote_gold};'> Coyote-thistle
    ↪ persists.</span>"
)

# writing subtitle
subtitle_text <- glue(
  "<span style='font-size:12pt; font-weight:400; color:{text_sub};'>",
  "Despite mirroring the trajectory of coyote-thistle's percent cover<br>",
  "at NCOS vernal pools, algae sees a major dropoff in 2024 and 2025",
  "</span>"
)
```

```

)

# defining caption text with data source
caption_text <- "Source: Cheadle Center for Biodiversity and Ecological
  ↪ Restoration"

### fonts ----
# Loading fonts directly with no dependency on utils/fonts.R
font_add_google("Oswald", "title_font")
font_add_google("Roboto", "text_font")
showtext_auto()
showtext_opts(dpi = 320)

# Creating a simple fonts list
fonts <- list(
  title = "title_font",
  text = "text_font"
)

```

5. Plot

```

### main plot ----
p <- filtered_explicit0 |>
  ggplot(aes(x = year, y = mean_pc_year, color = psoc, group = psoc)) +

  # Geom
  # Plotting horizontal lines
  geom_hline(
    yintercept = seq(0, 10, by = 2),
    color = grid_color,
    linewidth = 0.25
  ) +
  # plotting coyote-thistle line
  geom_line(
    data = filtered_explicit0 |> filter(psoc == "Eryngium_vaseyi"),
    linewidth = 1.9,
    alpha = 1.0
  ) +
  # plotting algae line
  geom_line(

```

```

    data = filtered_explicit0 |> filter(psoc == "algae"),
    linewidth = 1.3,
    alpha = 0.60
  ) +
  # plotting start point
  geom_point(
    data = filtered_explicit0 |> filter(year == 2021),
    size = 3.5,
    shape = 21,
    fill = bg_color,
    stroke = 1.8
  ) +
  # plotting end point
  geom_point(
    data = filtered_explicit0 |> filter(year == 2025),
    size = 3.5,
    shape = 21,
    fill = bg_color,
    stroke = 1.8
  ) +

  # Scales
  # assigning gold to coyote-thistle, green to algae
  scale_color_manual(
    values = c("Eryngium_vaseyi" = coyote_gold, "algae" = col_algae)
  ) +
  # customizing x axis limits
  scale_x_continuous(
    limits = c(2021, 2025),
    breaks = c(2021, 2022, 2023, 2024, 2025),
    expand = expansion(mult = c(0.03, 0.24))
  ) +
  # customizing y axis limits
  scale_y_continuous(
    limits = c(0, 10),
    breaks = seq(0, 10, by = 2),
    labels = \(x) glue("{x}%"),
    expand = expansion(mult = c(0.01, 0.04))
  ) +

  # Annotate
  # Adding dotted line at peak algae value
  annotate(

```

```

    "segment",
    x = algae_peak_year, xend = algae_peak_year,
    y = 0, yend = algae_peak_val,
    color = accent_coral,
    linewidth = 0.5,
    linetype = "dotted"
) +
# adding dotted line between endpoints
annotate(
    "segment",
    x = 2025, xend = 2025,
    y = 4.45, yend = 9.45,
    color = "#D8E4EE",
    linewidth = 0.45,
    linetype = "dotted"
) +
# adding floating caption
annotate(
    "text",
    x = 2025,
    y = 6.5,
    label = "5% difference\nin pct cover ",
    hjust = 0,
    size = 4,
    color = "#D8E4EE",
    family = fonts$text,
    lineheight = 1.1
) +
# adding floating caption for algae
annotate(
    "text",
    x = 2021.3,
    y = 1,
    label = glue("Algae"),
    hjust = 0,
    vjust = 0.5,
    size = 3.2,
    color = algae_green,
    family = fonts$text,
    lineheight = 1.15
) +
# adding floating caption for coyote-thistle
annotate(

```

```

    "text",
    x = 2021.3,
    y = 4,
    label = glue("Coyote-thistle"),
    hjust = 0,
    vjust = 0.5,
    size = 3.2,
    color = coyote_gold,
    family = fonts$text,
    lineheight = 1.15
  ) +
  # adding floating caption
  annotate(
    "text",
    x = 2022.7,
    y = 9,
    label = glue("Algae diverges from\n Coyote-thistle"),
    hjust = 0,
    vjust = 0.5,
    size = 3,
    color = accent_coral,
    family = fonts$text,
    lineheight = 1.15
  ) +

  # Labels
  labs(
    title = title_text,
    subtitle = subtitle_text,
    caption = caption_text,
    x = NULL,
    y = NULL
  ) +

  # Theme
  # using same theme and background aesthetics from the creator
  theme_void() +
  theme(
    # canvas
    plot.background = element_rect(fill = bg_color, color = NA),
    panel.background = element_rect(fill = bg_color, color = NA),

```



```

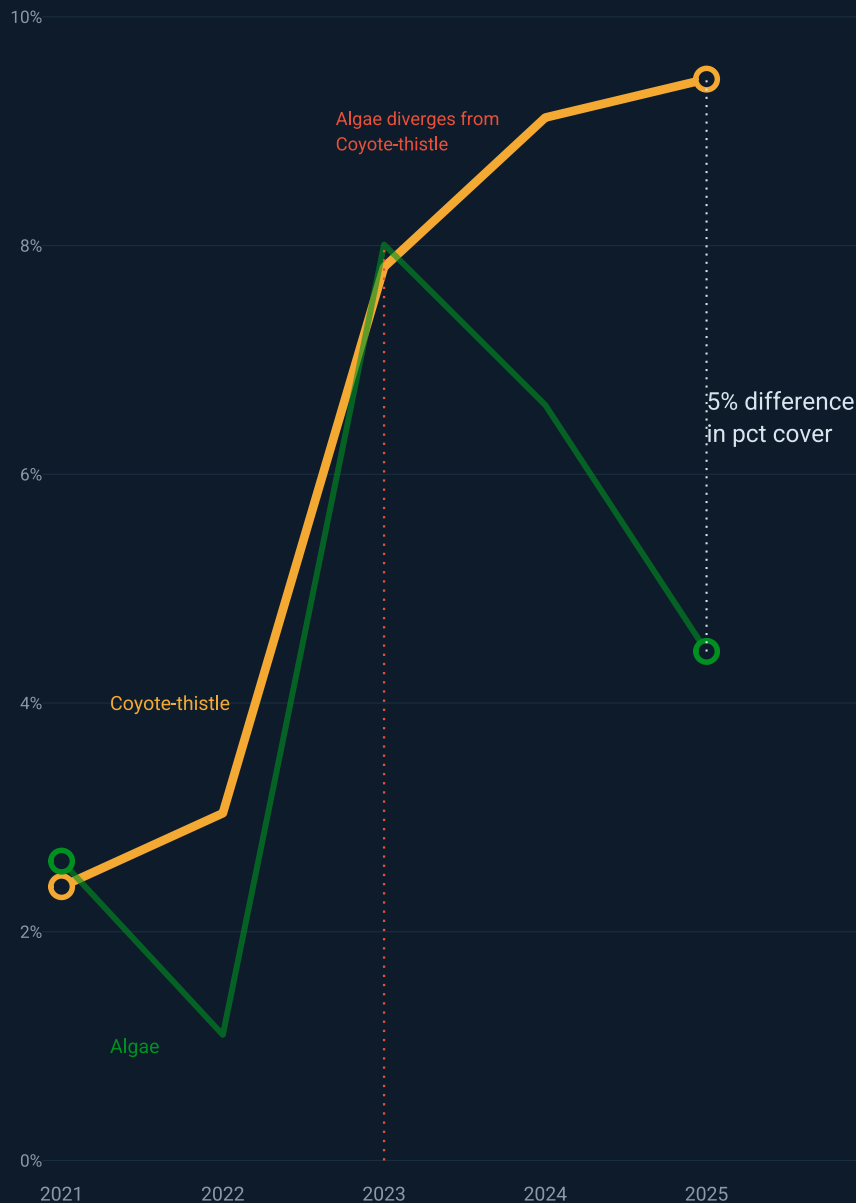
plot.title = element_markdown(
  family = fonts$title,
  size = 28,
  color = text_white,
  hjust = 0,
  lineheight = 1.1,
  margin = margin(t = 10, b = 6)
),
plot.subtitle = element_markdown(
  family = fonts$text,
  size = 10,
  color = text_sub,
  hjust = 0,
  lineheight = 1.4,
  margin = margin(b = 22)
),
plot.caption = element_markdown(
  family = fonts$text,
  size = 6.5,
  color = "#5A6878",
  hjust = 1,
  margin = margin(t = 14)
),
axis.text.x = element_text(
  family = fonts$text,
  size = 9,
  color = text_muted,
  margin = margin(t = 6)
),
axis.text.y = element_text(
  family = fonts$text,
  size = 8,
  color = text_muted,
  hjust = 1
),
axis.line = element_blank(),
axis.ticks = element_blank(),
panel.grid = element_blank(),
legend.position = "none",
plot.margin = margin(t = 20, r = 16, b = 16, l = 20)
)

```

p

Algae Falls. Coyote-thistle persists.

Despite mirroring the trajectory of coyote-thistle's percent cover at NCOS vernal pools, algae sees a major dropoff in 2024 and 2025



Source: Cheadle Center for Biodiversity and Ecological Restoration

4. Describing the Process

I chose to just copy and paste the code from the creators project website on github. Luckily the creator labeled every step of their process with clear and concise headers, making it fairly simple to follow along and grab the code chunks I needed. That said, the process of using someone else's code for my own visualization purposes was fairly difficult. Of course it was nice that there was a framework provided for me, and a clear example of what my final visualization would look like, but it was pretty troublesome to adjust the code to my own data. It took a much more than just swapping out data sets and labels - a lot of the user's code involved files that were integrated into their github website, so I had to find some workarounds, and truthfully AI was integral in troubleshooting errors. This visualization takes on a very different aesthetic theme compared to what we usually make for assignments. Firstly, the title is more visually striking, and it includes an informative subtitle which we've never included in class. Additionally, there are no axes titles, which is usually a must-have for our assignments, but the author of this visualization seems to believe that the subtitle and visuals are self explanatory enough, so I'm following the same philosophy for the sake of replication. The dark background is a fresh way to make the bright lines pop on on the chart, and the dashed lines are a cool new way to indicate important times in the data. Lastly, the ability to add captions on the plot was a cool new way to communicate the data.